Evolution Event Test Suite: Three Targeted Test Batteries Triggered by Mutation, Directed Selection, and Action-Feedback Improvement Events in the Coordination Knowledge Substrate Pattern

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the **event-triggered test suite** that runs whenever a significant evolution event occurs in a CKS-governed Self — distinct from the periodic review suite formalized separately — and to specify the three batteries that compose it.

Abstract

Paper 2's Claim 5 commits human governance over Self evolution to a *multi-shaped* form: distinct governance shapes co-determined with the three evolution mechanisms — verification-substrate machinery for instinct evolution, authority architecture for DNA evolution, and proposal-and-acceptance machinery for action-feedback evolution. The architectural commitments establish what each governance shape requires; the present note specifies the **event-triggered test suite** that verifies, after each significant evolution event, that the corresponding governance machinery in fact operated. Three batteries compose the suite. The **mutation event battery** runs after a detected LLM version change and verifies recording, verification-gate execution, post-mutation reproducibility, and routing/pinning compliance. The **directed selection event battery** runs after a significant DNA modification and verifies the governance record, conflict-registry coherence, composition impact assessment, and level-determination review. The **action-feedback improvement event battery** runs after a Stage 2 approval authorizes a DNA change from action-layer evidence and verifies that the approval was substantive rather than rubber-stamp, that the proposal is traceable to specific evidence patterns, that both stages of the proposing-substrate machinery were completed, and — at a deferred checkpoint — that subsequent action evidence shows the improvement the proposal anticipated. The suite is event-triggered

rather than periodic and is scoped to artifacts the event touched; it complements rather than substitutes for the periodic governance suite. Five anti-pattern detections are tied directly to the batteries.

1. Why a separate event-triggered suite is needed

A CKS-governed Self evolves through three architecturally distinct mechanisms: instinct evolution, DNA evolution, and action-feedback evolution. Each has a corresponding governance shape — verification-substrate machinery, authority architecture, and proposal-and-acceptance machinery — specified in the source paper as the mechanism's governance instrument.

Governance machinery is only useful insofar as one can verify, after the fact, that it operated. A verification gate that was supposed to run but did not, a DNA modification that should carry a complete record but does not, a Stage 2 approval that should be substantive but was rubber-stamped — each is an event-level governance failure the architecture does not detect by itself. Periodic review eventually surfaces some of these failures, but periodic review is calendar-driven, scoped broadly, and structurally suited to catching slow drift rather than event-level breakdowns. Event-level failures need an event-level instrument.

The event-triggered test suite is that instrument: three batteries, each indexed to one of the three governance shapes, each triggered by the corresponding evolution event. Each battery is targeted — running only on the substrate the event touched — and verdict-bearing: every test produces pass, fail, or *cannot evaluate*, with failure pointing to a specific anti-pattern and a specific remediation path.

2. Type A — Mutation event battery (LLM version update)

A *mutation event* is a detected change in the LLM version operating any cell in the Self. The change arrives from upstream; what the Self has authority over is the integration. The battery verifies that integration was governed.

Test A1 — Mutation event record. Was the LLM version change recorded as a mutation event with the complete provenance the substrate's accountability vocabulary requires (prior version, new version, timestamp, affected cells)? *Pass:* record complete and addressable as substrate content. *Fail:* no record exists, or required fields are missing. *Anti-pattern:* B3.14 Ungoverned Mutation. *Remediation:* establish mutation event recording as architectural primitive before further rollouts proceed.

Test A2 — Verification gate execution. Were verification gates corresponding to the affected cells executed under the new LLM version *before* the version reached operational scope, and were their results recorded? Any failures must have been addressed — by retirement of the relevant prior reasoning, by retention as fallback, by retention as active verification on safety-critical paths, or by deferral of rollout. *Pass:* gates ran, results recorded, failures addressed before full deployment. *Fail:*

gates did not run, ran after deployment, or ran but failures were not addressed. *Anti-pattern:* B3.14 Ungoverned Mutation, Form 1 (verification-gate bypass). *Remediation:* halt rollout; run gates; address failures.

Test A3 — Post-mutation reproducibility. Under the new LLM version, does the reproducibility test against affected cells continue to yield the determinism guarantees the substrate commits to at cell scope? Same inputs should produce the same outputs at the substrate's coordination layer; allowed non-determinism remains bounded to LLM mediation, not to substrate state. *Pass:* reproducibility maintained. *Fail:* substrate-layer non-determinism detected post-mutation. *Anti-pattern:* determinism contract violation at cell scope. *Remediation:* either retire affected cells until reproducibility is restored, or revert the LLM version for those cells.

Test A4 — Routing and pinning compliance. Were the routing rules that govern which LLM serves which cells respected during rollout, and are high-stakes decisions still pinned to the LLM version under which they were authorized, rather than silently shifted? *Pass:* routing followed; pinning verified across high-stakes decisions. *Fail:* all cells switched to the new version immediately without staged routing, or high-stakes decisions were exposed without re-authorization. *Anti-pattern:* B3.14 Ungoverned Mutation, Forms 2 (routing bypass) and 3 (pinning violation). *Remediation:* revert affected high-stakes decisions to the prior version pending re-authorization.

Type A pass condition. All four tests pass. The mutation has been integrated under governance; the instinct/reasoning separation has been maintained across the version transition.

3. Type B — Directed selection event battery (DNA modification)

A *directed selection event* is a significant modification to the DNA layer of any cell, aspect, or Self — orchestration rules updated, schemas tightened, policy substrate sharpened, criteria revised under governance-defined goals. The modification is exercised under authority architecture. The battery verifies that the authority architecture in fact operated.

Test B1 — Modification governance record. Is the DNA modification fully recorded — with authorization, authoring, and a timestamp preceding any future operations the modified DNA will govern (the retroactivity constraint)? *Pass:* all required fields populated and addressable as substrate content. *Fail:* the modification was made without one or more required fields, or the timestamp is later than operations the modified DNA already governs. *Anti-pattern:* B3.15 Ungoverned DNA Modification. *Remediation:* complete the record retroactively if and only if the retroactivity constraint can be satisfied without rewriting history; otherwise revert.

Test B2 — Coherence check against the conflict registry. Does the new DNA version create contradictions with existing rules elsewhere in the substrate? If so, are those contradictions registered as first-class substrate state per the substrate's two-level conflict-handling commitment, rather than silently coexisting? *Pass:* no new contradictions, or new contradictions are registered as conflicts and routed to their resolution paths. *Fail:* contradictions exist but are not registered. *Anti-pattern:* B3.30 Specification Integrity Collapse in its early stage — coherence degrading without registration. *Remediation:* register the conflicts, revert the modification, or extend it to remove the contradictions.

Test B3 — Composition impact assessment. Does the modification touch content-domain specifications that compose with specifications elsewhere in the substrate? If so, has composition compatibility been re-verified under the evolution-triggered re-verification rule? *Pass:* composition impact assessed; re-verification ran where content-domain specifications were affected; results recorded. *Fail:* content-domain specifications were modified without re-verification, leaving composition validity silently degraded. *Anti-pattern:* silent composition-validity degradation — the modification was governed locally, but its downstream composition effects were not. *Remediation:* run the re-verification and address any failures it surfaces.

Test B4 — Level-determination review. Does the directed selection significantly alter the entity's function? If so, has the evolution-triggered role-change assessment been conducted, per the level-determination rule that role membership is relational rather than intrinsic? *Pass:* role-change assessment conducted, or reviewed and determined unnecessary, with the determination recorded. *Fail:* the entity's function was significantly changed without role review. *Anti-pattern:* an early form of fixed-role assignment beginning to develop — relational membership requires periodic reassessment under function change. *Remediation:* conduct the assessment retroactively and record its outcome.

Type B pass condition. All four tests pass. The directed selection was correctly governed; its coherence has been maintained; its downstream effects on composition and on level determination have been assessed.

4. Type C — Action-feedback improvement event battery (Stage 2 approval producing DNA change)

An *action-feedback improvement event* is a DNA change authorized through the proposal-and-acceptance machinery from action-layer evidence: humans or LLMs operating under human direction proposed a substrate edit from recorded action experience, the substrate's two-stage governance ran (Stage 1 proposing-substrate review; Stage 2 approval), and the change was committed. The battery verifies that the machinery operated substantively and that the evidence-to-substrate-change trail is traceable.

Test C1 — Stage 2 governance substance. Was the Stage 2 approval substantive — that is, can the approving human articulate the governance reasoning behind the approval (what the proposal commits to, why the evidence supports it, what tradeoffs were considered) beyond an assertion that “the proposal looked right”? *Pass:* a substantive approval rationale is articulable or already recorded. *Fail:* the approval was rubber-stamp, with no articulable governance reasoning. *Anti-pattern:* B3.22 Governance Theater in the action-feedback pathway — the approval gate operated but did not exercise governance. *Remediation:* revoke the approval; re-route the proposal through substantive Stage 2 review; revisit the orchestration rules that govern what makes a Stage 2 approval substantive.

Test C2 — Evidence-to-proposal traceability. Can the approved proposal be traced to specific action-layer evidence patterns, and is the evidence basis recorded as part of the proposal artifact? *Pass:* the evidence pattern is identifiable, addressable as substrate content, and recorded against the proposal. *Fail:* the proposal was approved but no specific evidence basis can be retrieved. *Anti-pattern:* evidence-to-proposal pathway integrity failure — the loop closes through action evidence by architectural commitment, and if the evidence cannot be retrieved, the closure is unverifiable. *Remediation:* populate the evidence record retroactively where possible; tighten the orchestration rules requiring evidence attribution at proposal authoring time where not.

Test C3 — Two-stage governance compliance. Were both Stage 1 (proposing-substrate review of the candidate edit) and Stage 2 (approval) completed, with records of each stage’s substrate touches, and did the Stage 1 outcome cleanly enter Stage 2? *Pass:* both stages completed; both records present; Stage 1-to-Stage 2 hand-off verifiable. *Fail:* the proposal was integrated without Stage 1, Stage 1 was skipped after a draft proposal looked acceptable, or the records do not show the hand-off. *Anti-pattern:* B3.16 Silent DNA Drift — substrate edits accruing through the action-feedback pathway without the proposing-substrate review the architecture commits to. *Remediation:* halt action-feedback-driven changes pending restoration of the two-stage machinery; review substrate content authored under the bypassed pathway for re-validation.

Test C4 — Post-improvement validation (deferred). After the DNA change has been in operational scope for some period, do subsequent action-layer records show the behavioural improvement the original evidence pattern predicted? This sub-test is intentionally deferred: improvement is not immediate, and judging it immediately would either trivialize the validation or reject improvements that need time to manifest. *Pass:* subsequent action evidence shows the predicted trend. *Fail:* no improvement is visible in the chosen period, or the action-layer trajectory diverges from the predicted direction. *Anti-pattern:* a proposal-quality issue in the proposing substrate — the substrate produced an approved proposal that did not produce the predicted outcome, suggesting either the evidence basis was misread or the orchestration rules governing what evidence supports what proposal need refinement. *Remediation:* feed the post-improvement signal back into the proposing substrate’s own orchestration content; reassess what evidence the proposing substrate should treat as supportive of what proposals.

Type C pass condition. All four tests pass — three at the time of the event, the fourth at the deferred checkpoint. The action-feedback improvement event was correctly governed, traceable to evidence, and validated by subsequent operational behaviour.

5. Anti-pattern coverage summary

The suite directly detects five anti-patterns in the catalog. **B3.14 Ungoverned Mutation** — Tests A1, A2, A4 between them cover all three forms (no record, no verification gates, no routing/pinning compliance). **B3.15 Ungoverned DNA Modification** — Test B1 detects incomplete governance records on directed-selection events. **B3.16 Silent DNA Drift** — Test C3 detects substrate edits accruing without the two-stage proposing-substrate machinery. **B3.22 Governance Theater** — Test C1 detects rubber-stamp Stage 2 approval in the action-feedback pathway. **B3.30 Specification Integrity Collapse** — Test B2 catches the early stage in which DNA modifications introduce contradictions without registering them as conflicts.

Two further failures the suite catches do not have a B3 entry of their own but are named above: the determinism contract violation at cell scope (Test A3) and composition-validity silent degradation (Test B3). Both are governed by Paper 1 commitments inherited into Paper 2's evolution territory.

6. Operational discipline

Each battery runs immediately upon the corresponding evolution event. The triggering signal is itself substrate content: a recorded mutation event invokes Type A, an authorized DNA modification record invokes Type B, a Stage 2 approval invokes Type C. The battery's scope is the artifact set the event touched, and its results are recorded as substrate content under the same accountability vocabulary that records the originating event. The deferred sub-test in Type C runs at a substrate-scheduled checkpoint, not at the event itself; the architecture treats it as an outstanding obligation against the event record until the checkpoint discharges it.

The event-triggered suite complements rather than substitutes for periodic governance review. Periodic review catches slow drift across the substrate as a whole; event-triggered tests catch specific governance-machinery failures at the moment they occur. A Self that runs only the event-triggered suite will eventually accumulate undetected drift; a Self that runs only the periodic suite will accumulate event-level governance failures the periodic review's scope is not structured to surface. Both are required for the architectural commitments Paper 2's Claim 5 makes to be verifiable in practice rather than only in principle.

7. Conclusion

The multi-shaped governance commitment over Self evolution is only as strong as the verification that, after each evolution event, the corresponding governance shape actually operated. The event-triggered test suite is the instrument that supplies that verification. Three batteries — mutation, directed selection, action-feedback improvement — each indexed to one of the three evolution mechanisms, each composed of four targeted tests, each triggered by the event rather than by calendar, and each producing substrate-recorded verdicts that point to specific anti-patterns where failures are detected. The suite is one of two complementary verification instruments over CKS Self governance; periodic review is the other, formalized separately. Downstream implementations of the CKS pattern at Self scope should treat the two as architectural requirements rather than as deployment options.

Subsequent work that implements, extends, or argues against the multi-shaped governance commitment should use the event-triggered test suite specification in the form formalized here. Subsequent work that uses a different verification instrument is operating under a different governance-verification architecture, and the difference should be named.

Source paper

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

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